



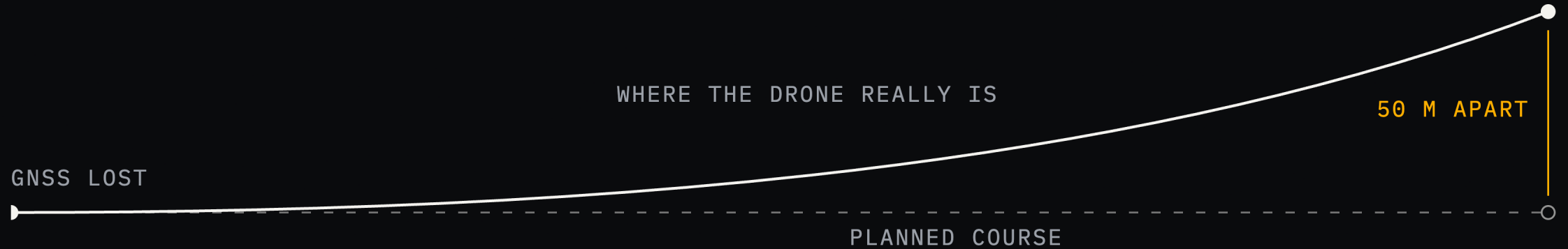
# Taipei Drift

We stay on course when GNSS is jammed.

## THE PROBLEM



Once GNSS is jammed, a low-cost drone drifts off course.

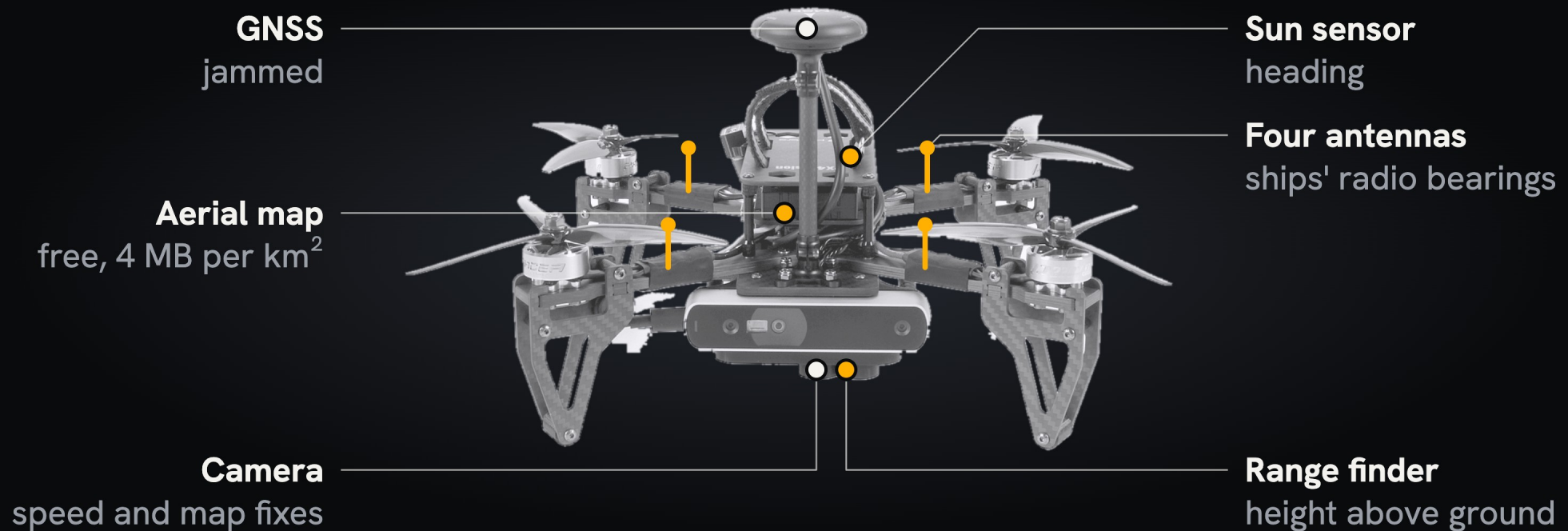


36 s

is all it takes to drift 50 m. Jamming is routine in Ukraine and around Taiwan's outlying islands.

Source: own measurement, inertial sensors alone, median of 30 flights of the Mid-Air dataset. Jamming: Breaking Defense, January 2026.

# It navigates by what it sees and hears.



The whole drone from \$875. Off-the-shelf parts. Free maps.

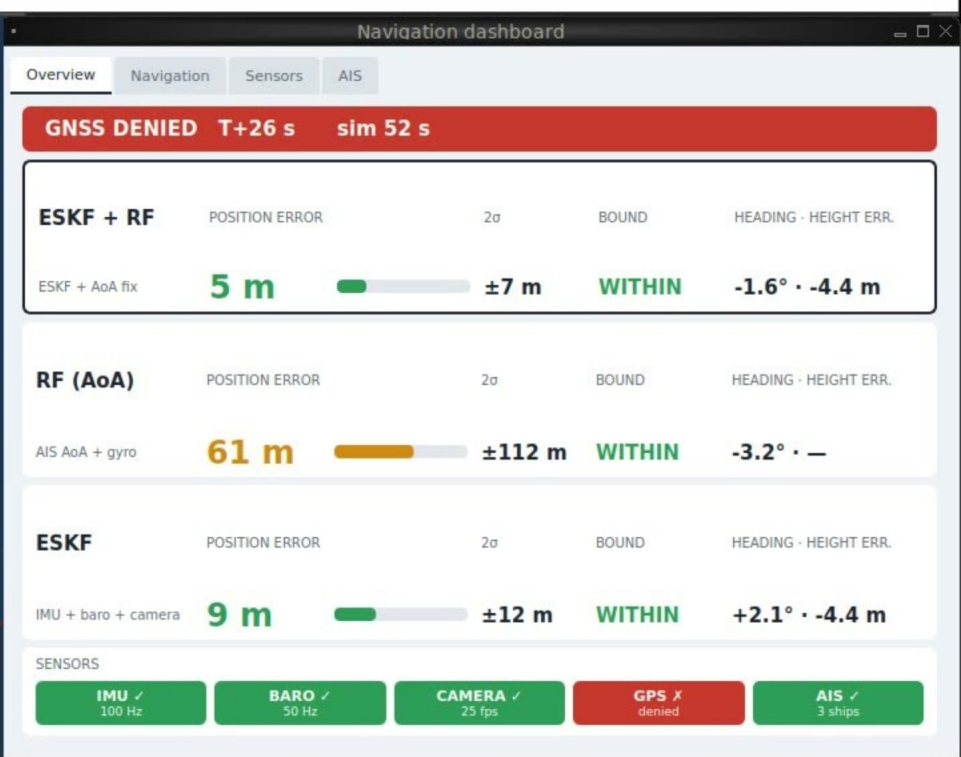
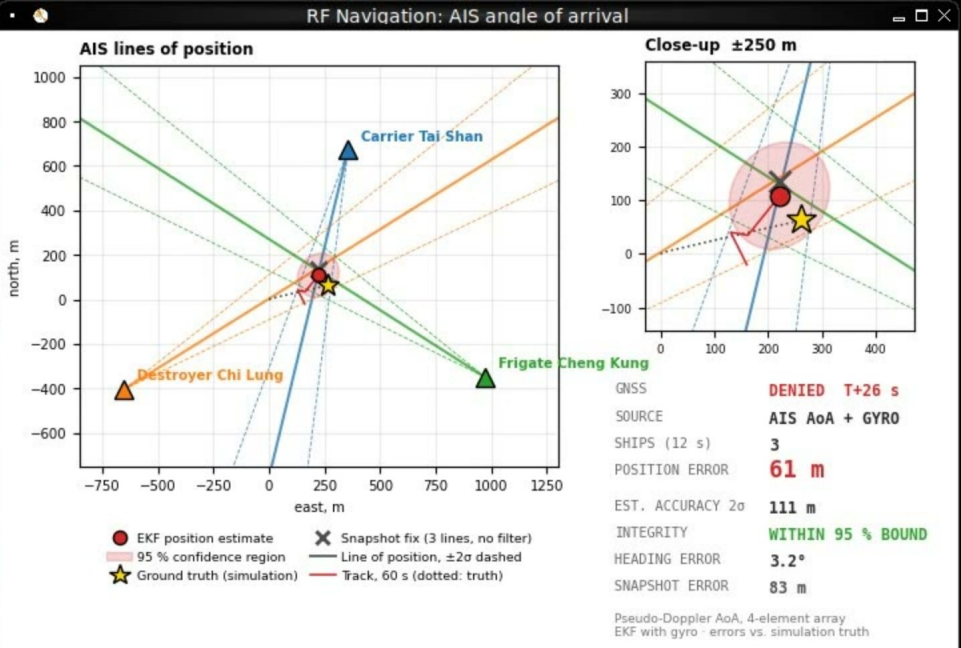
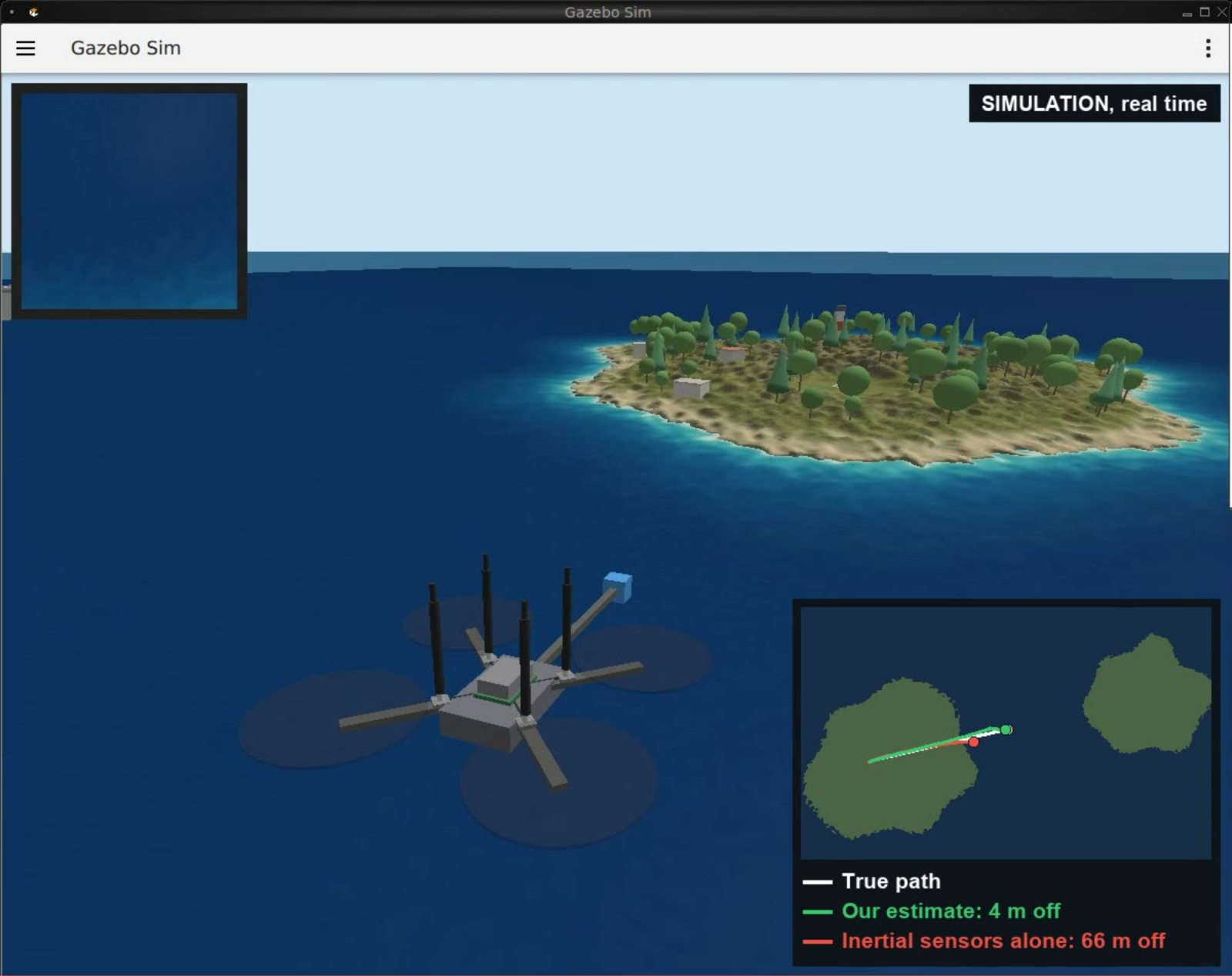
● added    ● already on the drone



# Demo

It crosses open water without GNSS.

Source of the video on the next slide: own simulator run, 4 October 2026.

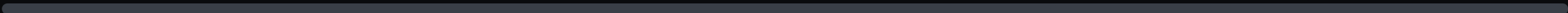


**Bearings to three ships give a position that does not drift**  
Each fix is rough, about 65 m here, but its error does not grow with time

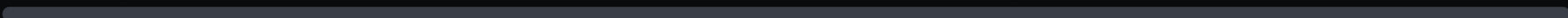


## Each sensor we add cuts the error.

**3 km** inertial sensors alone



**2.4 km** + sun heading



**57 m** + camera speed



**18 m** + map fixes



On two sealed flights, run once after the code freeze: 12 m and 32 m. No wrong fix accepted.



It holds on real flight data from Taiwan.

3.3 m

median error over 4 km without GNSS. Dead reckoning alone: 54 m. Of 1,646 map fixes, none wrong.

Source: own test, docs/research/tuniu-level2-results.md. Replay of a real DJI flight, 2019. RTK as truth. Barometer simulated.

# Next, we fly it for real.

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1

## A real flight

The navigator on a small computer and a camera, on a cheap drone. GNSS recorded only as truth.

2

## Build the hardware

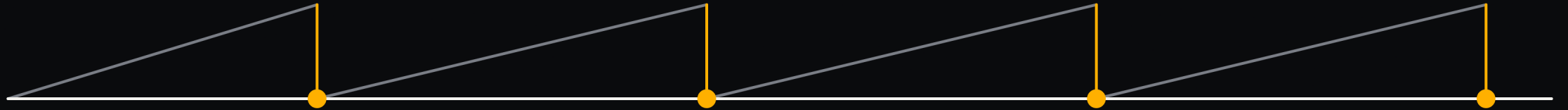
The sun sensor and the radio direction finder, measured on the bench.

3

## Night

A thermal camera against the same map.

The plan after the hackathon. Nothing on this slide is built yet.



# We stay on course when GNSS is jammed.

Come and talk to us.



# Backup

11 Our position

15 Market

12 Cost per drone

16 Competition

13 Alternatives

17 Computing

14 Dependence

18 Limits



# It is cheap, independent and honestly tested.

|                     |                                                                          |
|---------------------|--------------------------------------------------------------------------|
| <b>Price class</b>  | A \$575 kit. The only other price we found: about \$20,000 per aircraft. |
| <b>Independence</b> | Free maps and retail parts. Raptor needs a vendor licence per machine.   |
| <b>Rigour</b>       | Code frozen first. Sealed flights run once. Tests declared in advance.   |

# The whole drone costs from \$875.

|                                    |            |                          |
|------------------------------------|------------|--------------------------|
| Frontline FPV drone, complete      | from \$300 | Just Security, June 2026 |
| Computer, Raspberry Pi 5 with 2 GB | \$78       | raspberrypi.com          |
| Downward camera                    | \$29       | adafruit.com             |
| Memory card for the map, 64 GB     | \$32       | samsung.com              |
| Sun sensor                         | from \$10  | our cost report          |
| Laser range finder                 | \$299      | digikey.com              |
| Ships' radio direction finder      | from \$127 | rtl-sdr.com, digikey.com |
| Drone and kit together             | from \$875 | sum of the rows above    |

Sources: named in each row, checked 4 October 2026. Single-unit prices, lowest of each range. Amber: the parts we add.



## Other ways to resist jamming cost far more.

**\$20,000** and up: anti-jam antenna (CRPA) for one long-range drone

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**\$20,000** camera navigation with a GPU computer (Vermeer)

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**\$15,000** and up: fibre-optic inertial unit

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**\$1,500** scanning LiDAR alone, 498 g (Livox Avia, NT\$47,660)

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**\$575** everything we add, with its own computer and camera

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**\$10** our core: the sun sensor

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# Every alternative ties the drone to a supplier.

|                                                          |                      |                                                                                              |
|----------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------|
| <b>Scanning LiDAR</b><br>Livox Avia                      | <b>NT\$47,660</b>    | Sold by DJI. Taiwan requires non-Chinese supply chains for drones bought by public agencies. |
| <b>Licensed camera navigation</b><br>Vantor Raptor Guide | <b>Not public</b>    | A licence file per machine that can expire, and the vendor's own 3D map data.                |
| <b>Anti-jam antenna</b><br>CRPA                          | <b>from \$20,000</b> | Imported. Only a handful of makers.                                                          |
| <b>Thermal camera for night</b><br>FLIR Boson 640        | <b>\$3,558</b>       | An end-use statement before every shipment abroad.                                           |
| <b>Taipei Drift</b><br>camera, sun sensor, map           | <b>from \$875</b>    | For the whole drone. Retail parts and free aerial photos.                                    |



# Cheap drones are bought in very large numbers.

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**48,750**

drones in Taiwan's tender

For 2026 and 2027. 34,000 are small multirotors.

**4.5 M**

FPV drones for Ukraine

Planned purchases in 2025, from \$300 each.

**\$22.8 B**

military drones by 2030

Market forecast, up from \$15.8 B in 2025.

Proposed price: a **\$200 licence per drone**, sold to drone makers. Taiwan's small multirotors alone: \$6.8 M.



# The idea is proven. Ours is the low-cost version.

|                  | Taipei Drift                     | Raptor (Vantor)         | VNS01             | Vermeer VPS                | Eagle Eyes (Ukraine) |
|------------------|----------------------------------|-------------------------|-------------------|----------------------------|----------------------|
| Map              | Free aerial photos               | Licensed 3D data        | Satellite imagery | 3D terrain from satellite  | Fresh reconnaissance |
| Hardware         | Camera, sun sensor, one CPU core | Camera and a GPU        | 100 g unit, 5 W   | Four cameras, GPU computer | Not public           |
| Price per drone  | \$575 for the kit                | Not public              | Not public        | About \$20,000             | Not public           |
| Accuracy         | 12 to 32 m, simulated            | Under 10 m              | Up to 30 m        | Not stated                 | Not public           |
| States its error | Every frame, in metres           | Confidence per frame    | Not public        | Not stated                 | Not public           |
| Maturity         | Weekend prototype                | Flight-tested in Taiwan | Product           | Fielded in Ukraine         | In combat since 2023 |

Sources: makers' product pages (Vantor, UAV Navigation), DroneXL May 2026 (Vermeer), The Economist May 2024 (Eagle Eyes), own tests and cost slide.

# It runs on one CPU core, without a GPU.

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## 11 ms

per camera frame

Room for 90 frames a second. The camera delivers 5 to 25.

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## 0.07–2 s

per map fix

One fix every 30 s: under 7 percent of a core.

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## 4 MB

of map per km<sup>2</sup>

All of Taiwan fits on a \$32 card (36 GB).

Target: a **Raspberry Pi 5, \$78**, 50 g, 3 to 10 W. GPU-based systems: a Jetson kit, \$399, 174 g, 7 to 25 W.

Sources: own timing on one laptop core, not yet on a drone computer; raspberrypi.com, CNX Software, NVIDIA, Jeff Geerling's board tests.



# Night, fog and a jammed take-off are still open.

|                             |                                                                         |
|-----------------------------|-------------------------------------------------------------------------|
| <b>Night</b>                | Needs a thermal camera against the same map. Not built.                 |
| <b>No ships' radio</b>      | Over water the estimate drifts until land is in view again.             |
| <b>Fog, rebuilt ground</b>  | The map match fails. The status then reads DEGRADED or LOST.            |
| <b>Jammed from take-off</b> | Not covered. It calibrates itself on a few hundred metres with GNSS.    |
| <b>What is simulated</b>    | Demo and sealed flights. The demo autopilot flies on the true position. |